

Lab Assignment 02



Inspiring Excellence

Course Code:	CSE111
Course Title:	Programming Language II
Topic:	OOP Basics, Instance Variable and Instance Method
Number of Tasks:	11 (Coding: 08, Tracing: 03)

[Submit all the Coding Tasks (Task 1 to 8) in the Google Form shared on buX before the next lab. Submit the Tracing Tasks (Task 9 to 11) handwritten to your Lab Instructors at the beginning of the lab]

[You are not allowed to change the driver codes of any of the tasks]

Task 1

You are given the following “**University**” class:

```
public class University{  
    public String name;  
    public String country;  
}
```

Now write a Java **tester** class named “**UniversityTester**”.

- Write the main method and create 2 objects of **University** class and print the location of the objects and print the instance variables of the objects. Are the location of the objects the same?
- Now change the instance variables of the first object.
name = “Imperial College London”
country = “England”

Now change the instance variables of the second object.

name = “BRAC University”
country = “Bangladesh”

Now check if the instance variables of both objects have changed or not and whether the instance variables of both objects are of the same value or not.

Task 2

Design the “**Student**” class so that the main method prints the following:

Tester Class	Output
<pre>public class StudentTester1{ public static void main(String [] args){ Student s1 = new Student(); System.out.println("Name of the Student: "+s1.name); System.out.println("ID of the Student: "+s1.id); s1.name = "Bob"; s1.id = 123; System.out.println("Name of the Student: "+s1.name); System.out.println("ID of the Student: "+s1.id); } }</pre>	<pre>Name of the Student: Default ID of the Student: 0 Name of the Student: Bob ID of the Student: 123</pre>

Task 3

Design the **CSECourse** class to generate the correct output from the driver code provided below:

Driver Code	Output
<pre>public class CourseTester{ public static void main(String args []){ CSECourse c1 = new CSECourse(); System.out.println("Course Name: "+c1.courseName); System.out.println("Course Code: "+c1.courseCode); System.out.println("Credit: "+c1.credit); } }</pre>	Course Name: Programming Language II Course Code: CSE111 Credit: 3

Task 4

Design the “**ImaginaryNumber**” class to generate the **output** given below:

Tester Class	Output
<pre>public class Tester6{ public static void main(String [] args){ ImaginaryNumber num1 = new ImaginaryNumber(); String p = num1.printNumber(); System.out.println(p); System.out.println("1*****"); num1.realPart=3; num1.imaginaryPart=7; System.out.println(num1.printNumber()); System.out.println("2*****"); ImaginaryNumber num2 = new ImaginaryNumber(); num2.realPart=1; num2.imaginaryPart=9; System.out.println(num2.printNumber()); } }</pre>	0 + 0i 1***** 3 + 7i 2***** 1 + 9i

Task 5

Design the **Course** class to generate the correct output from the driver code provided below:

Driver Code	Output
<pre>public class Tester5{ public static void main(String[] args) { Course c1 = new Course(); Course c2 = new Course(); c1.updateDetails("Programming Language I","CSE110", 3); System.out.println("===== 1 ====="); c1.displayCourse(); c2.updateDetails("Data Structures","CSE220", 3); System.out.println("===== 2 ====="); c2.displayCourse(); c1.updateDetails("Programming Language II","CSE111", 3); System.out.println("===== 3 ====="); c1.displayCourse(); } }</pre>	<pre>===== 1 ===== Course Name: Programming Language I Course Code: CSE110 Course Credit: 3 ===== 2 ===== Course Name: Data Structures Course Code: CSE220 Course Credit: 3 ===== 3 ===== Course Name: Programming Language II Course Code: CSE111 Course Credit: 3</pre>

Task 6

Implement the “**Assignment**” class with necessary properties, so that the given output is produced for the provided driver code.

Driver Class	Output
<pre>public class AssignmentTester{ public static void main(String [] args){ Assignment as1 = new Assignment(); as1.printDetails(); System.out.println("1-----"); as1.tasks = 11; as1.difficulty = "Moderate"; as1.submission = true; as1.printDetails(); System.out.println("2-----"); System.out.println(as1.makeOptional()); System.out.println("3-----"); as1.printDetails(); System.out.println("4-----"); Assignment as2 = new Assignment(); as2.tasks = 12; as2.difficulty = "Hard"; as2.submission = false; as2.printDetails(); System.out.println("5-----"); System.out.println(as2.makeOptional()); } }</pre>	<pre>Number of tasks: 0 Difficulty level: null Submission required: false 1----- Number of tasks: 11 Difficulty level: Moderate Submission required: true 2----- Assignment will not require submission 3----- Number of tasks: 11 Difficulty level: Moderate Submission required: false 4----- Number of tasks: 12 Difficulty level: Hard Submission required: false 5----- Submission is already not required</pre>

Task 7

Design the **CellPhone** class so that the **main** method of tester class can produce the following output:

Tester Class	Output
<pre>public class Tester9{ public static void main(String[]args){ CellPhone phone1 = new CellPhone(); phone1.printDetails(); phone1.model ="Nokia 1100"; System.out.println("1#####"); phone1.storeContact("Joy - 01834"); System.out.println("====="); phone1.printDetails(); System.out.println("2#####"); phone1.storeContact("Toya - 01334"); phone1.storeContact("Aayan - 01135"); System.out.println("====="); phone1.printDetails(); System.out.println("3#####"); phone1.storeContact("Sani - 01441"); System.out.println("====="); phone1.printDetails(); } }</pre>	<pre>Phone Model unknown Contacts Stored 0 1##### Contact Stored ===== Phone Model Nokia 1100 Contacts Stored 1 Stored Contacts: Joy - 01834 2##### Contact Stored Contact Stored ===== Phone Model Nokia 1100 Contacts Stored 3 Stored Contacts: Joy - 01834 Toya - 01334 Aayan - 01135 3##### Memory full. New contact can't be stored. ===== Phone Model Nokia 1100 Contacts Stored 3 Stored Contacts: Joy - 01834 Toya - 01334 Aayan - 01135</pre>

Task 8

Create an **Employee** class to provide the expected output.

- An employee will have a name, salary and designation.

- The name will be assigned inside the newEmployee() method
- Whenever a New Employee joins his/her salary will be **Tk. 30,000** and the designation will be **junior**.
- Employees with salaries greater than **Tk. 50,000** and **Tk. 30,000** need to pay **30%** and **10%** of salary as tax respectively.
- Employees can be promoted to **senior**, **lead** and **manager** positions. Based on their promotion they will get an increment of **Tk. 25,000**, **Tk. 50,000** and **Tk. 75,000** respectively.

Driver Code	Expected Output
<pre> public class Tester3{ public static void main(String[] args){ Employee emp1 = new Employee(); Employee emp2 = new Employee(); Employee emp3 = new Employee(); emp1.newEmployee("Harry Potter"); emp2.newEmployee("Hermione Granger"); emp3.newEmployee("Ron Weasley"); System.out.println("1 ====="); emp1.displayInfo(); System.out.println("2 ====="); emp2.displayInfo(); System.out.println("3 ====="); emp3.displayInfo(); System.out.println("4 ====="); emp1.calculateTax(); System.out.println("5 ====="); emp1.promoteEmployee("lead"); System.out.println("6 ====="); emp1.calculateTax(); System.out.println("7 ====="); emp1.displayInfo(); System.out.println("8 ====="); emp3.promoteEmployee("manager"); System.out.println("9 ====="); emp3.calculateTax(); System.out.println("10 ====="); emp3.displayInfo(); } } </pre>	<pre> 1 ===== Employee Name: Harry Potter Employee Salary: 30000.0 Tk Employee Designation: junior 2 ===== Employee Name: Hermione Granger Employee Salary: 30000.0 Tk Employee Designation: junior 3 ===== Employee Name: Ron Weasley Employee Salary: 30000.0 Tk Employee Designation: junior 4 ===== No need to pay tax 5 ===== Harry Potter has been promoted to lead New Salary: 80000.00 Tk 6 ===== Harry Potter Tax Amount: 24000.0 Tk 7 ===== Employee Name: Harry Potter Employee Salary: 80000.0 Tk Employee Designation: lead 8 ===== Ron Weasley has been promoted to manager New Salary: 105000.00 Tk 9 ===== Ron Weasley Tax Amount: 31500.0 Tk 10 ===== Employee Name: Ron Weasley Employee Salary: 105000.0 Tk Employee Designation: manager </pre>

Task 9

Consider the following class:

```
public class Human{
    public int age;
    public double height;
}
```

Show the output of the following sequence of statements:

Human h1 = new Human();	Output
Human h2 = new Human();	
h1.age = 21;	
h1.height = 5.5;	
System.out.println(h1.age);	
System.out.println(h1.height);	
h2.height = h1.height - 3;	
System.out.println(h2.height);	
h2.age = h1.age++;	
System.out.println(h1.age);	
h2 = h1;	
System.out.println(h2.age);	
System.out.println(h2.height);	
h2.age++;	
h2.height++;	
System.out.println(h1.age);	
System.out.println(h1.height);	
h1.age++ = ++h2.age;	
System.out.println(h2.age);	
System.out.println(h2.height);	

Task 10

Consider the following class:

```
public class Student{
    public String name;
    public double cgpa;
}
```

Show the output of the following sequence of statements:

[illegible]

Task 11

1	<code>public class Task11 {</code>
2	<code> public int p = 3, y = 2, sum;</code>
3	<code> public void methodA(){</code>
4	<code> int x = 0, y = 0;</code>
5	<code> y = y + this.y;</code>
6	<code> x = sum + 2 + p;</code>
7	<code> sum = x + y + methodB(p, y);</code>
8	<code> System.out.println(x + " " + y+ " " + sum);</code>
9	<code> }</code>
10	<code> public int methodB(int p, int n){</code>
11	<code> int x = 0;</code>
12	<code> y = y + (++p);</code>
13	<code> x = x + 2 + n;</code>
14	<code> sum = sum + x + y;</code>
15	<code> System.out.println(x + " " + y+ " " + sum);</code>
16	<code> return sum;</code>
17	<code> }</code>
18	<code>}</code>

Driver code:

<pre> public class Tester11 { public static void main(String [] args){ Task11 t1 = new Task11(); t1.methodA(); t1.methodA(); } } </pre>	Outputs		
	x	y	Sum

Ungraded Tasks (Optional)

(You don't have to submit the ungraded tasks)

Task 1

Complete the “Cat” class so the main method produces the following output:

Test Class	Output
<pre>public class Test7{ public static void main(String [] args){ Cat c1 = new Cat(); System.out.println("====="); c1.printCat(); c1.color = "Black"; System.out.println("====="); c1.printCat(); c1.color = "Brown"; c1.action = "jumping"; System.out.println("====="); c1.printCat(); } }</pre>	<pre>===== White cat is sitting ===== Black cat is sitting ===== Brown cat is jumping</pre>

Task 2

Complete the **Bird** class so that main method produces the following **output**:

Test class	Output
<pre>public class Test8{ public static void main(String args[]) { Bird b1 = new Bird(); b1.name = "Parrot"; b1.flyUp(3); b1.makeNoise(); b1.flyDown(5); b1.flyDown(2); b1.flyDown(1); Bird b2 = new Bird(); b2.name = "Eagle"; b2.flyUp(5); } }</pre>	<pre>Parrot has flown up 3 feet. Squawk Parrot cannot fly down 5 feet. Parrot has flown down 2 feet. Parrot has flown down 1 feet and landed. Eagle has flown up 5 feet. Eagle has flown down 5 feet and landed. Squee</pre>

<pre> b2.flyDown(5); b2.makeNoise(); } }</pre>	
--	--

Task 3

Implement the “**ChickenBurger**” class with necessary properties, so that the given output is produced for the provided driver code.

[**Note:**

1. There are four available spice levels: **Mild**, **Spicy**, **Naga** and **Extreme**. You can store these values in a String array.
2. You might need to use the `.equals()` method to compare two string values.]

Driver Class	Output
<pre> public class BurgerMaker{ public static void main(String [] args){ ChickenBurger b1 = new ChickenBurger(); System.out.println(b1.bun); System.out.println(b1.price); System.out.println(b1.sauceOption); System.out.println(b1.spiceLevel); System.out.println("-----1-----"); System.out.println(b1.serveBurger()); System.out.println("-----2-----"); b1.customizeSpiceLevel("Extreme Jhaal"); b1.customizeSpiceLevel("Spicy"); System.out.println("-----3-----"); System.out.println(b1.serveBurger()); System.out.println("-----4-----"); ChickenBurger b2 = new ChickenBurger(); b2.bun = "Brioche"; b2.price += 50; b2.sauceOption = "Regular";</pre>	<pre> Sesame 200 Less Not Set -----1----- Cannot serve now. Customize Spice Level first. -----2----- This spice level is unavailable. Spice level set to Spicy. -----3----- The burger is being served:- Bun Type: Sesame Price: 200 Sauce Option: Less Spice Level: Spicy -----4----- Spice level set to Naga. -----5----- The burger is being served:- Bun Type: Brioche Price: 250 Sauce Option: Regular Spice Level: Naga</pre>

<pre> b2.customizeSpiceLevel("Naga"); System.out.println("-----5-----"); System.out.println(b2.serveBurger()); } </pre>	
---	--

Task 4

Implement the “**MobilePhone**” class with necessary properties to produce the given output for the provided driver code.

Driver Class	Output
<pre> public class MobilePhoneTester{ public static void main(String args []){ MobilePhone m1 = new MobilePhone(); MobilePhone m2 = new MobilePhone(); m1.setContactCapacity(5); m2.setContactCapacity(100); m1.details(); System.out.println("1-----"); m1.addContact("John", 9866); m1.addContact("Maria", 7865); System.out.println("2-----"); m1.details(); System.out.println("3-----"); m1.makeCall(9866); System.out.println("4-----"); m1.addContact("Henry", 2365); System.out.println("5-----"); m1.makeCall(7552); m1.makeCall(2365); System.out.println("6-----"); m1.addContact("Gomes", 4589); m1.addContact("Antony", 8421); m1.addContact("Tony", 5789); System.out.println("7-----"); m1.details(); } } </pre>	<pre> Total Contacts: 0 Contact List: 1----- The contact of John is added. The contact of Maria is added. 2----- Total Contacts: 2 Contact List: John:9866 Maria:7865 3----- Calling John . . . 4----- The contact of Henry is added. 5----- Calling 7552 . . . Calling Henry . . . 6----- The contact of Gomes is added. The contact of Antony is added. Storage Full!! 7----- Total Contacts: 5 Contact List: John:9866 Maria:7865 Henry:2365 Gomes:4589 Antony:8421 </pre>

--	--

Task 5

1	public class Test1{
2	public int sum;
3	public int y;
4	public void methodA(){
5	int x=2, y =3;
6	int [] msg ={3, 7};
7	y = this.y + msg[0];
8	methodB(msg[1]++, msg[0]);
9	x = x + this.y + msg[1];
10	sum = x + y + msg[0];
11	System.out.println(x + " " + y+ " " + sum);
12	}
13	public void methodB(int mg2, int mg1){
14	int x = 0;
15	y = this.y + mg2;
16	x = x + 19 + mg1;
17	sum = this.sum + x + y;
18	mg2 = y + mg1;
19	mg1 = mg2 + x + 2;
20	System.out.println(x + " " + y+ " " + sum);

21	}
22	}

<pre>public class Tester5{ public static void main (String args[]){ Test1 t1 = new Test1(); t1.methodB(5,-8); Test1 t2 = new Test1(); t2.methodA(); } }</pre>	Outputs		